NF=10 / NF=1 Noise Simulation Combined Report

Codex analysis

2026-05-15 HKT

Executive Summary

Both raw result sets are captured and archived: NF=10 / noisescale=10 sampling-noise stress mode and NF=1 / noisescale=1 full-noise mode. Each data set contains 20 completed points across four t_d settings and five process corners. All Spectre points completed with 0 errors. The analysis below is corner-by-corner; averages are not used as the primary conclusion.

Noise Calculation

Assume $V_{FS} = V_{ref} = 1$ Vpp and 12 bits:

$$
\begin{aligned}
LSB &= V_{FS}/2^{12} = 244.141\,\mu V \\
V_{q,rms} &= LSB/\sqrt{12} = 70.477\,\mu V \\
P_q &= 4.967054 \times 10^{-9}\,V^2 \\
P_{sig} &= (V_{FS}/(2\sqrt{2}))^2 = 0.125000\,V^2 \\
P_{total} &= P_{sig}/10^{SNR/10} \\
P_{excess} &= \max(P_{total} - P_q, 0)
\end{aligned}
$$

If $V_{FS} = 2V_{ref}$ is used, RMS values scale by 2 and powers by 4; t_d trends are unchanged.

Table 1: Best t_d by corner using SNR as the selection criterion.

Corner	NF10 t_d	NF10 SNR	NF10 ENOB	NF1 t_d	NF1 SNR	NF1 ENOB
Nominal	27n	64.24	10.33	25n	68.44	11.10
C0	26n	64.40	10.42	25n	69.55	11.20
C1	27n	61.54	9.95	25n	67.80	10.92
fs	26n	64.86	10.46	25n	69.12	11.16
sf	27n	61.83	9.92	27n	67.93	11.03

Figures

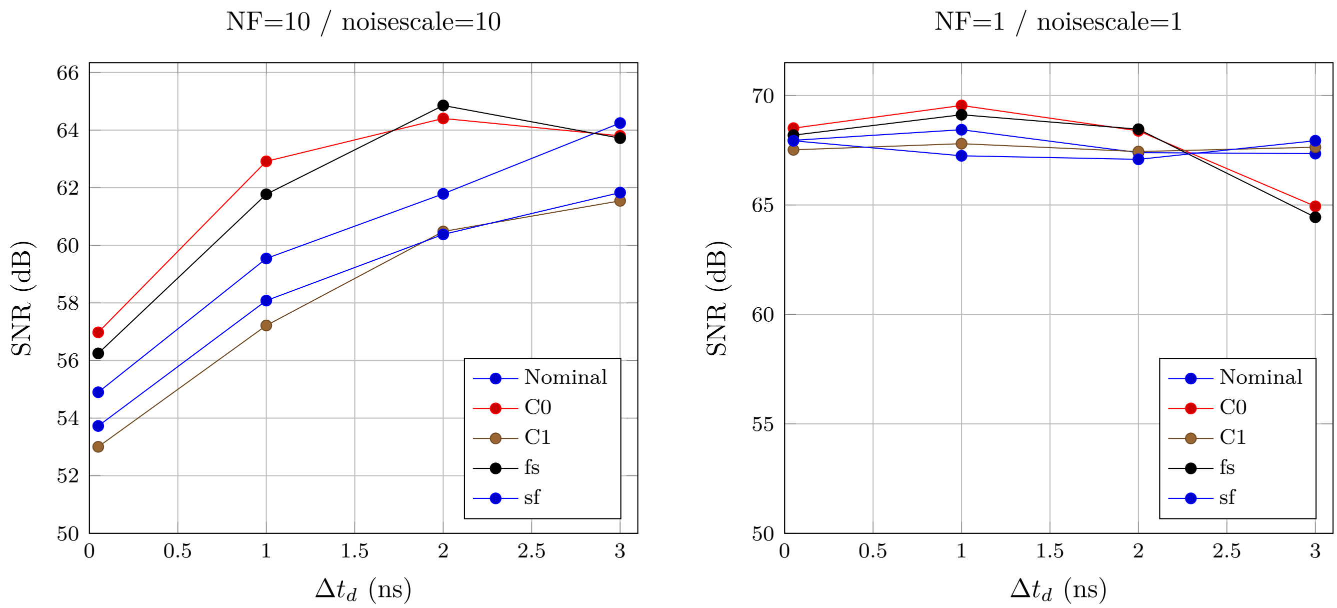

Figure 1: SNR versus Δt_d by corner.

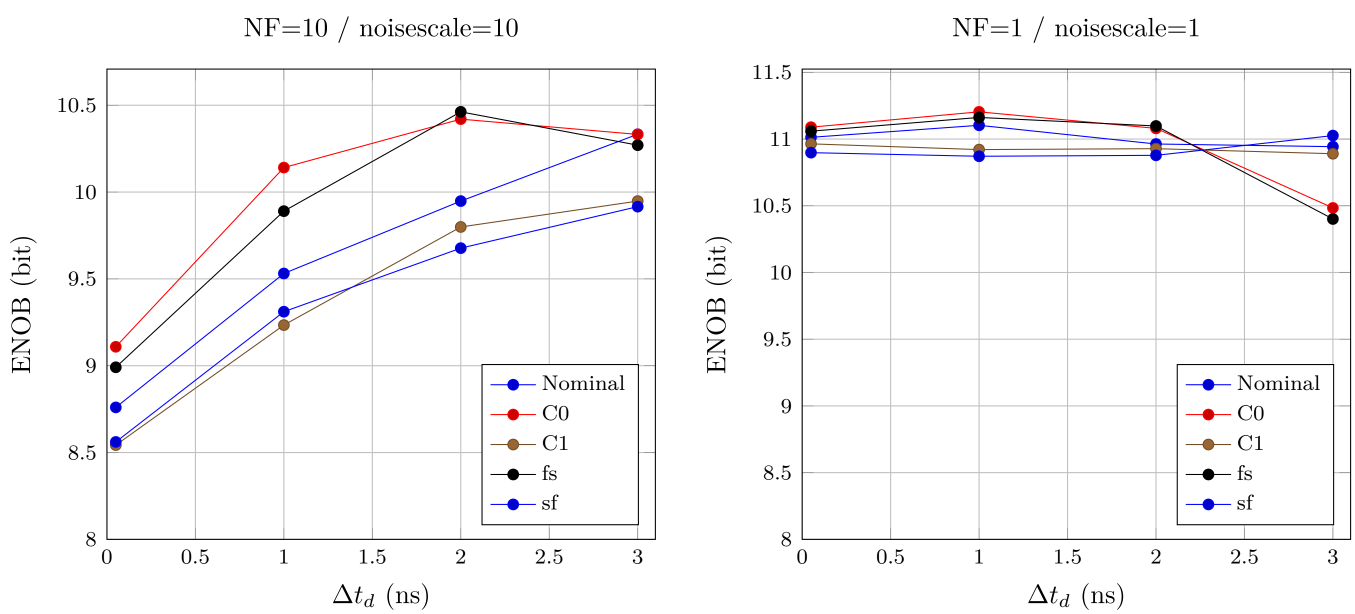

Figure 2: ENOB versus Δt_d by corner.

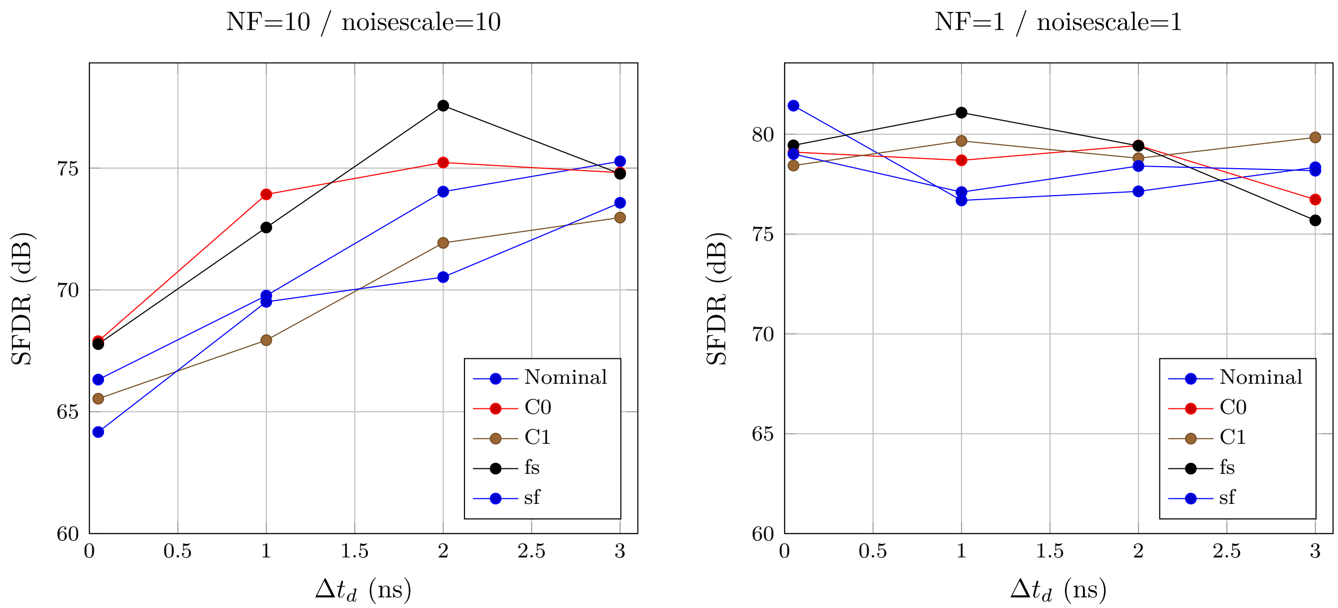

Figure 3: SFDR versus Δt_d by corner.

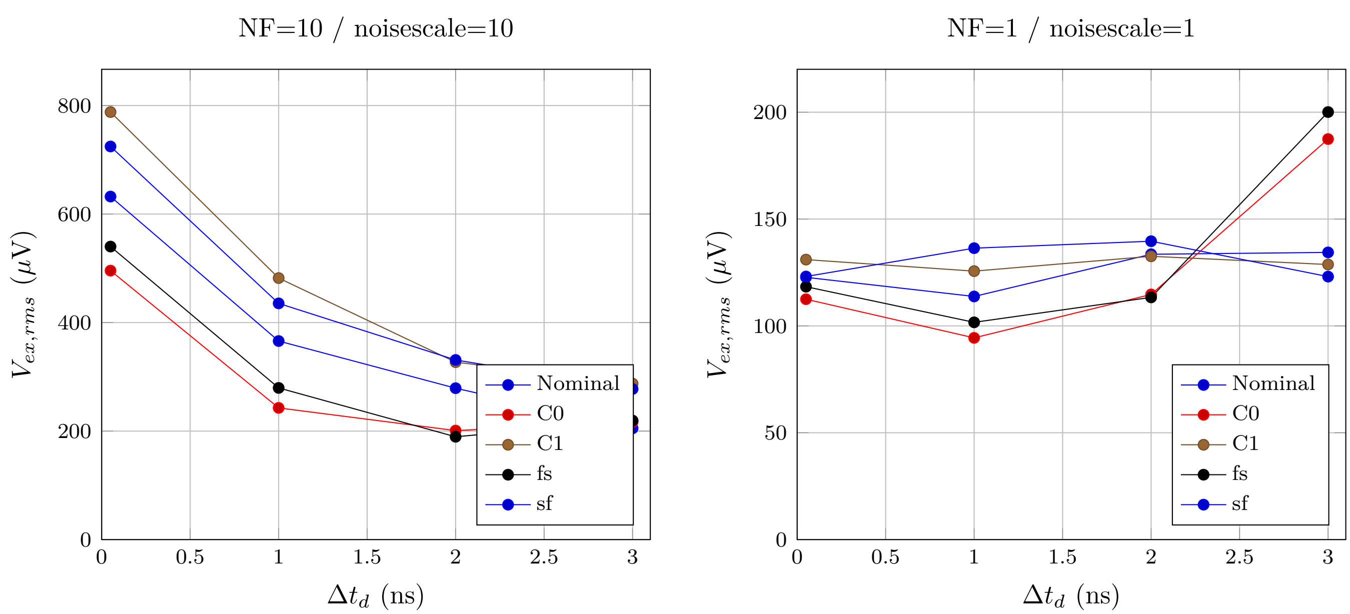

Figure 4: Inferred non-quantization noise RMS versus Δt_d by corner.

Corner-by-Corner Data

Nominal

t_d	Δt_d ns	NF10 SNR	NF10 ENOB	NF10 SFDR	NF10 V_{ex} uV	NF1 SNR	NF1 ENOB	NF1 SFDR	NF1 V_{ex} uV	ΔSNR
24.05n	0.05	54.90	8.76	66.32	632.2	67.95	11.01	81.43	122.7	13.05
25n	1.00	59.54	9.53	69.76	366.0	68.44	11.10	76.69	113.8	8.90
26n	2.00	61.79	9.95	74.03	279.0	67.39	10.96	77.14	133.5	5.60
27n	3.00	64.24	10.33	75.28	205.1	67.35	10.94	78.35	134.4	3.10

C0

t_d	Δt_d ns	NF10 SNR	NF10 ENOB	NF10 SFDR	NF10 V_{ex} uV	NF1 SNR	NF1 ENOB	NF1 SFDR	NF1 V_{ex} uV	ΔSNR
24.05n	0.05	56.98	9.11	67.90	495.7	68.51	11.09	79.10	112.5	11.53
25n	1.00	62.91	10.14	73.92	242.7	69.55	11.20	78.70	94.4	6.63
26n	2.00	64.40	10.42	75.23	201.0	68.38	11.08	79.44	114.7	3.98
27n	3.00	63.80	10.33	74.81	217.1	64.94	10.48	76.73	187.4	1.14

C1

t_d	Δt_d ns	NF10 SNR	NF10 ENOB	NF10 SFDR	NF10 V_{ex} uV	NF1 SNR	NF1 ENOB	NF1 SFDR	NF1 V_{ex} uV	ΔSNR
24.05n	0.05	53.00	8.54	65.54	788.0	67.52	10.96	78.43	131.0	14.51
25n	1.00	57.22	9.23	67.94	482.0	67.80	10.92	79.67	125.6	10.58
26n	2.00	60.48	9.80	71.93	327.0	67.44	10.93	78.80	132.6	6.96
27n	3.00	61.54	9.95	72.97	287.5	67.64	10.89	79.84	128.7	6.09

fs

t_d	Δt_d ns	NF10 SNR	NF10 ENOB	NF10 SFDR	NF10 V_{ex} uV	NF1 SNR	NF1 ENOB	NF1 SFDR	NF1 V_{ex} uV	ΔSNR
24.05n	0.05	56.25	8.99	67.78	540.1	68.19	11.06	79.44	118.3	11.94
25n	1.00	61.77	9.89	72.56	279.5	69.12	11.16	81.08	101.7	7.35
26n	2.00	64.86	10.46	77.56	189.5	68.46	11.10	79.42	113.3	3.61
27n	3.00	63.72	10.27	74.77	219.3	64.44	10.40	75.69	200.1	0.72

sf

t_d	Δt_d ns	NF10 SNR	NF10 ENOB	NF10 SFDR	NF10 V_{ex} uV	NF1 SNR	NF1 ENOB	NF1 SFDR	NF1 V_{ex} uV	ΔSNR
24.05n	0.05	53.73	8.56	64.17	724.6	67.93	10.90	79.01	123.1	14.20
25n	1.00	58.08	9.31	69.51	435.4	67.25	10.87	77.10	136.4	9.17
26n	2.00	60.38	9.68	70.52	331.1	67.08	10.88	78.41	139.6	6.71
27n	3.00	61.83	9.92	73.58	277.6	67.93	11.03	78.18	123.1	6.10

Engineering Readout

NF=10 stress mode confirms that t_d is a meaningful sampling-noise cancellation knob. The optimum is corner-dependent: Nominal, C1, and sf favor 27 ns by SNR, while C0 and fs favor 26 ns. NF=1 full-noise mode is cleaner in absolute SNR because it uses noisescale=1, while NF=10 intentionally stresses the selected sampling-noise contribution. Use the limiting corner, not an average, to choose the production t_d target.